Miscalibration Without Tradable Alpha: Displayed Prices, Actionable Quotes, and Fees in a Regulated Prediction Market

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Abstract

Prediction-market prices are widely read as probability forecasts, but calibration and economic exploitability are different estimands. We joined Kalshi snapshots to resolutions in overlapping strata—937 prospectively observed markets (331 event clusters) and 57,808 markets reconstructed from historical order-book candles (19,718 clusters)—and moved from displayed prices or midpoints to executable quotes and taker returns. Event-clustered bootstrap inference showed that retrospective midpoints were numerically close to perfect calibration but distinguishable at this scale (gap −1.98 percentage points, 95% CI [−2.51, −1.43]; slope 1.038 [1.017, 1.059]). Residual gaps were localized mid-curve, concentrated in wide books, increased with time to resolution, and were larger at displayed prices. Under the standard-fee scenario, all 20 bucket-direction intervals were below zero; pooled expectancy remained negative with zero fees. No displayed coherence violation was executable at contemporaneous quotes. A displayed probability is a forecast; an actionable quote is an offer.

Keywords: probability forecasting; calibration; prediction markets; market microstructure; favorite–longshot bias; limits to arbitrage

1. Introduction

In 2026, anyone with an internet connection could watch a federally regulated exchange display the probability that inflation would exceed 4%, that the Federal Reserve would cut rates, or that a specific team would win a specific game—and, in principle, bet against any number that looked wrong. Prediction-market prices increasingly circulate as public probabilities: journalists quote them, researchers aggregate them, and exchanges present them as evidence of forecasting performance. A contract displayed at 96 cents is routinely paraphrased as a 96% chance.

That convention merges two claims that ought to be kept apart. The first is epistemic: the reported price is calibrated, so events assigned 96% occur about 96% of the time. The second is economic: if that price were wrong, a trader could profit by correcting it. The second claim is commonly used to justify the first, but the claims rest on different price objects. Calibration studies often evaluate a displayed price, last trade, closing price, or midpoint. A taker transacts at a side-specific quote and may pay a fee. Whether an apparent calibration gap survives that transformation is an empirical question—and a different question from whether the displayed number was calibrated.

This paper estimates both quantities on the same venue and, within each measurement stratum, at the same entry policy. We move sequentially from the displayed price or order-book midpoint to the actionable ask and then to the fee-adjusted return from holding one contract to settlement. We also test whether displayed violations of logical coherence among related contracts were executable at contemporaneous quotes. The contribution is not a claim that prediction markets are broadly wrong. It is a measurement argument: the price object is part of the estimand.

1.1. Prior evidence and the price-object problem

Prediction-market prices are informative and often compare favorably with alternative forecasts (Wolfers & Zitzewitz, 2004), including over long election horizons (Berg et al., 2008). This literature ordinarily treats a transaction or reference price as the forecast and asks how well it predicts an eventual outcome. Even a frictionless price need not equal traders' mean belief (Manski, 2006); price-as-probability is an interpretive convention before it is an empirical claim. Page and Clemen (2013) moved closer to the present question: they documented horizon-dependent miscalibration in Intrade prices and showed that apparent excess returns were sensitive to the time value of locked capital. Grant et al. (2019) formalized the same intuition as a cost of capital. Our wedge is directly observed in the order book: the distance between the reported price and the quote available to a taker, plus fees.

A related literature documents the favorite–longshot pattern in betting and prediction markets: longshots are overpriced and favorites underpriced (Snowberg & Wolfers, 2010). Berg and Rietz (2019) report related evidence from the Iowa Electronic Market, and Bürgi et al. (2026) show that low-priced Kalshi contracts underperform while high-priced contracts do better. Those studies evaluate transaction-level or book-adjacent prices. In our prospective sample, the upper tail initially points in the opposite direction at the displayed price. The apparent contradiction attenuates when the analysis moves from the display to the order book—and at full scale, the midpoint curve takes the classic favorite–longshot shape.

Microstructure research supplies the economic discipline. Options studies test put–call parity against executable bid–ask bounds rather than midquotes (Ofek et al., 2004), and the limits-to-arbitrage framework emphasizes that a pricing discrepancy is not self-correcting merely because it is visible (Shleifer & Vishny, 1997). Recent prediction-market studies make the same move: Gebele et al. (2026) distinguish payoff-space violations from protocol-executable trades, Cheng et al. (2026) find that logically detectable arbitrage in high-frequency order-book data is rare, fleeting, and depth-constrained, and Saguillo et al. (2025) trace realized arbitrage in on-chain markets. Kalshi's own research reports strong aggregate calibration near resolution (Kagan & Baiocchi, 2026). We do not dispute that result; our contribution concerns horizons away from resolution and, more fundamentally, the difference between the platform's displayed signal and the trades available against it.

1.2. The present research

We report results from three data strata and one pre-specified decision rule. The prospective stratum comprises daily snapshots of an economics-focused feed of open markets joined to resolutions (937 markets, 331 event clusters; July 12–August 20, 2026). The retrospective stratum reconstructs an entry observation for settled markets from the exchange's public candlestick history (57,808 markets, 19,718 clusters; entry observations from August 2025 through August 2026). A coherence stratum scans, up to three times daily, for displayed monotonicity violations among threshold ladders and tests whether each was executable at actionable quotes. Before the tradability analysis, we committed a decision rule to the project's version-controlled governance record: a price bucket would count as economically exploitable only with $n \geq 200$ resolved observations, at least 30 event clusters, and a cluster-bootstrap 95% interval for fee-adjusted net expectancy entirely above zero.

2. Institutional setting and data

2.1. Kalshi and fee scenarios

Kalshi is a U.S. Commodity Futures Trading Commission–regulated designated contract market offering binary contracts that settle at $1.00 if the specified event occurs and $0 otherwise. Trading occurs in a central limit order book with maker and taker roles. The exchange's standard taker-fee formula is, in cents per contract, the ceiling of $0.07 \times P \times (1 - P) \times 100$, where P is the transaction price in dollars. Public fee materials also identify series with different or no fees (Kalshi, 2026a, 2026b). Historical market-specific fee schedules are not publicly archived. We therefore treat 0.07 as a standard-fee scenario—not as the reconstructed fee for every observation—and report pooled sensitivity over coefficients from 0 (the literal frictionless lower bound) to 0.14. Settlement itself carries no fee.

2.2. Prospective snapshot stratum

From July 12, 2026, an automated pipeline recorded up to three daily snapshots of eligible open markets: ticker, displayed price, best bid, best ask, volume, open interest, and hours to close. The feed initially covered an economics-focused universe of roughly 2,100 markets per day and excluded machine-generated multivariate combination contracts. A full-exchange broadening began on August 24, after the prospective analysis freeze; the repository release data-freeze-v1 followed on August 26. Accordingly, the prospective results characterize the eligible pre-freeze feed, not every open market on the exchange.

A separate job queried snapshotted markets after scheduled close and recorded the official YES/NO settlement. The predefined entry policy used each market's earliest observed snapshot. This is an early-entry policy, not inherently a conservative one: its bias and precision may differ from later-horizon policies. Markets without a two-sided book at entry contributed to displayed-price calibration but not to quote-based economics. The frozen sample contained 937 resolved markets in 331 exchange-defined event clusters from July 12 through August 20, 2026.

2.3. Retrospective reconstruction stratum

For historical markets, the pipeline reconstructed an entry observation from Kalshi's public candlestick endpoint, which reports open, high, low, and close values for best bids and asks. For each settled binary market with a lifetime of at least 6 hours, excluding sub-hourly index and weather churn, the analysis selected the earliest daily candle with a two-sided book and retained its closing bid and ask. The midpoint, actionable quotes, and fee-adjusted returns are therefore the authoritative retrospective stages; any historical display proxy is secondary.

At the analysis freeze (git tag data-freeze-v1, August 26, 2026), 57,808 markets in 19,718 event clusters had a qualifying two-sided early book and constituted the retrospective analysis set. Entry observations span August 2025 through August 2026. Because this window overlaps the prospective period (770 markets appear in both strata) and imposes a two-sided-book condition at entry, the retrospective stratum is a scale and construction check—not an independent out-of-sample replication. The strata are analyzed separately and their market or cluster counts are not summed.

2.4. Coherence scans

Within a threshold ladder, $P(X > t_1) \geq P(X > t_2)$ whenever $t_1 < t_2$. Within an exhaustive partition, component probabilities should sum to one. Up to three times daily, the instrument recorded displayed violations of ladder monotonicity and tested whether the corresponding two-leg position was available at contemporaneous quotes with positive locked profit after fees. For partitions, only the short-side test was treated as actionable because incomplete bucket groupings can create spurious long-side violations.

2.5. Estimands and inference

Calibration is summarized two ways. First, in 10-cent stage-specific price buckets: the observed settlement rate, the mean implied probability, and event-clustered bootstrap percentile intervals. Second, by direct event-clustered bootstrap of the calibration gap $g = \text{mean}(Y \times 100 - p)$, the Brier score (Brier, 1950), and the linear calibration slope and intercept of settlement on price (perfect calibration: slope 1, intercept 0). The economic estimand is mean cents per contract from the relevant side-specific taker entry, held to settlement and net of the assumed fee scenario.

The main analyses used 10,000 bootstrap draws with replacement at the exchange-defined event level and a fixed random seed. The fee-sensitivity intervals used 2,000 event-clustered draws. Bucketwise intervals are not familywise-error adjusted and are interpreted descriptively; the pooled global estimates are the confirmatory economics. The pre-specified exploitability rule requires a positive interval and therefore becomes harder, not easier, to satisfy under a familywise correction.

2.6. Open practices and computational assistance

The collection pipeline, analysis scripts, datasets, and version-controlled governance record are preserved in a frozen release identified as data-freeze-v1, with a crosswalk from each exhibit to the script and data object that generated it. An analysis-grade export of the frozen release (data-freeze-v1) — datasets, analysis scripts, the frozen artifacts, and the post-freeze robustness script — is publicly archived at https://github.com/ken10677/mwta-kalshi-replication; the full collection pipeline and version-controlled governance record are available from the author. The repository's commit history records the exploitability rule before the tradability analyses.

Generative AI tools assisted research-code development. All generated code was reviewed, version-controlled, and checked against the frozen outputs; broader manuscript-preparation uses are disclosed in the declaration before the references.

3. Results

3.1. The prospective display gap attenuated at the midpoint

Across 937 resolved markets, displayed prices implied approximately 312 YES resolutions; 257 occurred (global display gap −5.86 percentage points, 95% CI [−9.23, −2.53]; calibration slope 0.871 [0.774, 0.967]). The discrepancy was concentrated in the tails and ran in one direction: each display-price bucket resolved YES at or below its mean implied rate. In the 90–99-cent bucket, 84.0% resolved YES despite a 96.4% mean display (n = 144; CI [75.9, 91.4]). In the 0–9-cent bucket, the corresponding values were 2.0% and 3.2% (n = 403; CI [0.8, 3.5]).

On the 583-market subsample with two-sided books, the upper-tail gap was substantially smaller at the midpoint. Markets with 90–99-cent midpoints resolved YES 90.4% of the time against a 94.8% mean midpoint (n = 73; CI [81.9, 97.3]). The evidence is consistent with stale or otherwise unrepresentative displays on thin books, but the design does not identify staleness as the mechanism. The correction was not uniform: in the 40–49-cent midpoint bucket, 19.0% resolved YES against a 45.5% mean midpoint (n = 42; CI [8.1, 32.5]). Table 1 places these estimates beside the retrospective forecast metrics. Brier scores should not be compared across strata because their base rates and market composition differ.

Table 1
Global forecast metrics by stratum and price object

Stratum / object	n	Gap, pp [95% CI]	Brier [95% CI]	Slope [95% CI]	Intercept [95% CI]
Retrospective display	57,808	−2.43 [−2.95, −1.91]	.2005 [.1984, .2027]	.986 [.965, 1.006]	−.018 [−.028, −.007]
Retrospective midpoint	57,808	−1.98 [−2.51, −1.43]	.1999 [.1979, .2020]	1.038 [1.017, 1.059]	−.037 [−.048, −.026]
Prospective display	937	−5.86 [−9.23, −2.53]	.1049 [.0869, .1251]	.871 [.774, .967]	−.016 [−.035, .004]
Prospective midpoint	583	−7.50 [−12.07, −2.89]	.1413 [.1220, .1641]	.920 [.790, 1.032]	−.042 [−.079, −.004]

Note. Gap = mean(Y × 100 − p), in percentage points. CIs use event-clustered bootstrap resampling. Brier scores are not comparable across strata with different outcome base rates and market composition.

3.2. No prospective bucket met the exploitability gate

At contemporaneous quotes under the coded standard-fee scenario, no prospective price bucket met the pre-specified exploitability rule. The strongest NO-side candidates were positive but small and imprecise: 20–29 cents, +6.98 cents per contract (95% CI [−2.44, 15.27]; n = 56); 30–39 cents, +6.60 [−4.77, 16.86] (n = 35); and 80–89 cents, +7.93 [−8.84, 25.07] (n = 42). Each fell short of the required sample size and each interval included zero.

The mirror trade supplied a directional check. Buying YES at the ask had reliably negative expectancy in 8 of 10 buckets; in the 0–9-cent bucket the estimate was −16.37 cents per contract (95% CI [−22.47, −10.82]). Buying NO against 90–99-cent displays returned −13.44 cents (95% CI [−26.33, 0.23]). These results are not equivalence tests. They establish that the prospective data did not satisfy the pre-specified criterion for actionable alpha under the evaluated taker strategy.

3.3. At scale: close aggregate calibration with a localized favorite–longshot curve

The retrospective reconstruction (57,808 markets; 19,718 event clusters) was numerically close to calibrated at the order-book midpoint, although the sample was large enough to distinguish the global estimates from perfect calibration: the midpoint gap was −1.98 percentage points (95% CI [−2.51, −1.43]), with slope 1.038 [1.017, 1.059] and intercept −0.037 [−0.048, −0.026]. The corresponding display-price figures were a gap of −2.43 [−2.95, −1.91] and slope 0.986 [0.965, 1.006]. The interesting structure was not merely global error but its location.

Table 2 and Figure 1 report the complete midpoint calibration curve. The tails were close to calibrated: 0–9-cent midpoints resolved YES 5.4% of the time against 6.4% implied (CI [4.5, 6.4]), and 90–99-cent midpoints 94.0% against 93.1% (CI [92.7, 95.4]). The middle of the curve was systematically YES-overpriced: 30–39-cent midpoints resolved 30.5% against 35.5% implied (n = 7,920; CI [29.2, 31.8]), 40–49-cent midpoints 43.5% against 46.3% implied (n = 15,253; CI [42.6, 44.3]), and 50–59-cent midpoints 50.1% against 52.9% implied (n = 9,480; CI [48.8, 51.5]). At the favorite end the sign reversed: 70–79-cent midpoints resolved 75.4% against 74.4% implied, and 80–89-cent midpoints 85.7% against 84.5% implied (CI [84.2, 87.1]). At book-adjacent prices and at scale, the pattern was the classic favorite–longshot curve — though Section 3.7 shows this residual is carried almost entirely by wide-spread books.

The displayed-price 90–99-cent bucket, by contrast, showed 88.5% observed against 93.6% implied (n = 1,845; CI [86.7, 90.2])—a display-layer gap of roughly five points that was absent at the midpoint. This difference is consistent with, but does not identify, stale reference values on thin books.

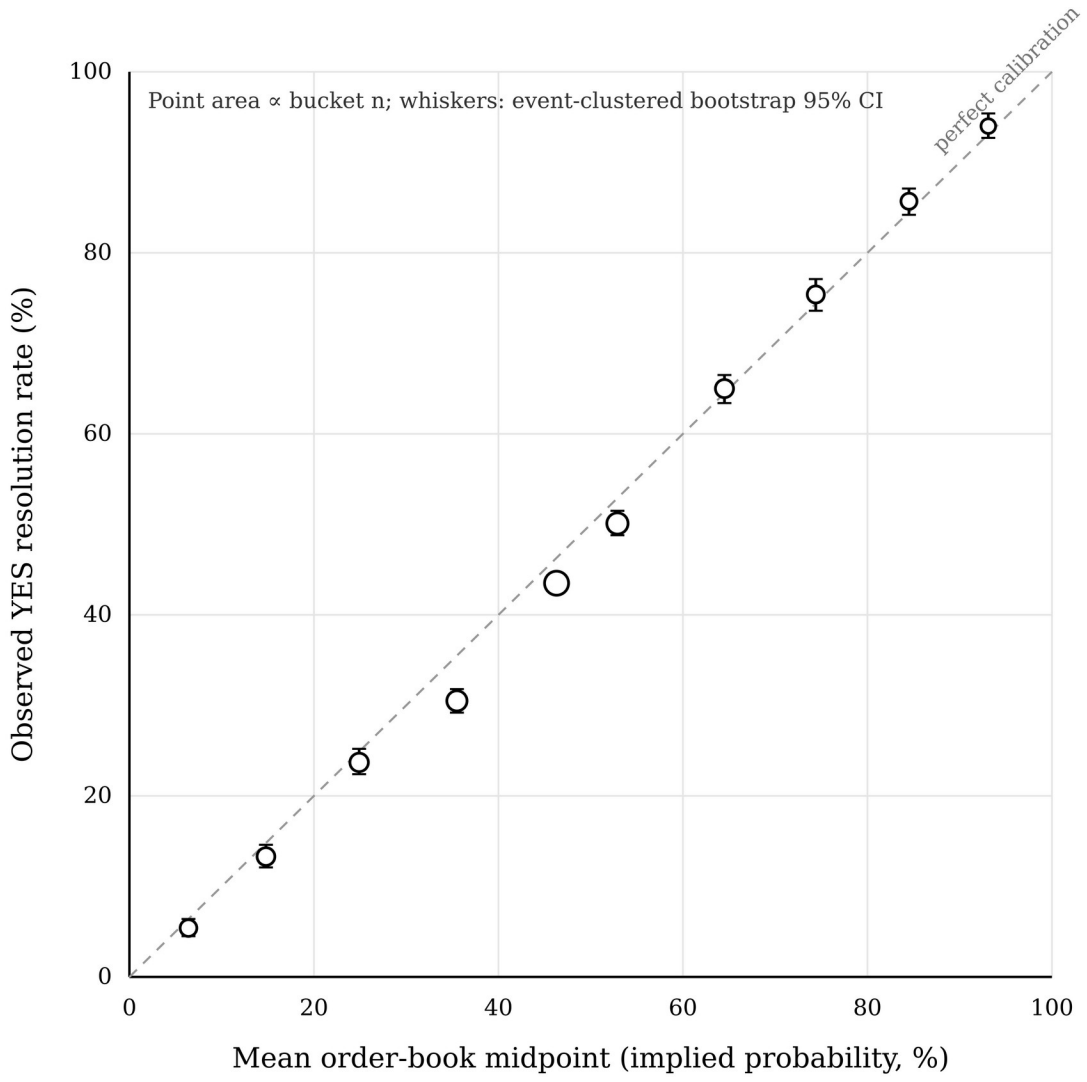
Table 2

Retrospective calibration at the order-book midpoint

Midpoint bucket (¢)	n	Clusters	Mean implied	YES rate	95% CI
0–9	3,255	1,744	6.4%	5.4%	[4.5, 6.4]
10–19	4,345	2,839	14.8%	13.3%	[12.1, 14.6]
20–29	5,206	3,698	24.9%	23.7%	[22.4, 25.2]
30–39	7,920	5,002	35.5%	30.5%	[29.2, 31.8]
40–49	15,253	9,241	46.3%	43.5%	[42.6, 44.3]
50–59	9,480	6,005	52.9%	50.1%	[48.8, 51.5]
60–69	4,745	3,981	64.5%	65.0%	[63.4, 66.5]
70–79	3,768	2,958	74.4%	75.4%	[73.6, 77.1]
80–89	2,458	2,082	84.5%	85.7%	[84.2, 87.1]
90–99	1,378	1,158	93.1%	94.0%	[92.7, 95.4]

Note. Intervals are event-clustered bootstrap 95% CIs for the YES resolution rate. Cluster counts do not sum because an event can contribute markets to more than one bucket.

Figure 1
Reliability of retrospective order-book midpoints



Note. The diagonal marks perfect calibration. Point area is proportional to bucket n; whiskers show event-clustered bootstrap 95% CIs.

3.4. Taker economics: negative under the standard-fee scenario in every bucket

The economic estimand did not inherit the calibration pattern. Under the standard 0.07 fee scenario, all 20 bucket-direction cells (10 buckets × buy-YES/buy-NO) had 95% intervals entirely below zero (Table 3). Buying YES at the ask lost between −3.76 cents per contract (80–89 cents; CI [−5.26, −2.25]) and −30.12 cents (50–59 cents; CI [−31.55, −28.68]). Buying NO lost between −2.37 cents (0–9 cents; CI [−3.40, −1.42]) and −25.00 cents (50–59 cents; CI [−26.19, −23.78])—including the mid-curve buckets in which the calibration gap favored NO, because the spread exceeded the three-to-five-point bias.

A preliminary positive cell at the 90–99-cent YES ask (+3.30 cents at n = 10 in an early batch) reversed as the sample grew: at n = 1,845 the estimate was −5.29 cents (95% CI [−6.94, −3.69]). This is the kind of small-sample result the pre-specified gate was designed not to elevate.

Table 3

Retrospective taker returns under the standard-fee scenario

Bucket (¢)	Buy NO at ask, ¢/contract [95% CI]	Buy YES at ask, ¢/contract [95% CI]
0–9	−2.37 [−3.40, −1.42]	−13.60 [−15.44, −11.74]
10–19	−2.60 [−3.80, −1.44]	−8.95 [−10.48, −7.46]
20–29	−6.38 [−7.81, −4.97]	−9.15 [−10.68, −7.61]
30–39	−10.53 [−11.80, −9.27]	−20.70 [−22.29, −19.06]
40–49	−24.42 [−25.36, −23.51]	−28.63 [−29.63, −27.64]
50–59	−25.00 [−26.19, −23.78]	−30.12 [−31.55, −28.68]
60–69	−7.04 [−8.71, −5.37]	−8.59 [−10.22, −6.95]
70–79	−8.19 [−9.99, −6.37]	−6.58 [−8.35, −4.81]
80–89	−10.78 [−12.35, −9.19]	−3.76 [−5.26, −2.25]
90–99	−12.72 [−14.47, −10.94]	−5.29 [−6.94, −3.69]

Note. The 0.07 coefficient is a standard-fee scenario, not a claim about each market's historical fee. Intervals use 10,000 event-clustered bootstrap draws.

The pooled global estimates were negative under every fee coefficient examined (Table 4). Under the standard 0.07 scenario, pooled buy-NO expectancy was −14.52 cents per contract (95% CI [−15.08, −13.96]) and pooled buy-YES expectancy was −18.57 [−19.24, −17.87]. Setting fees to zero changed neither sign nor interval conclusion: −12.99 [−13.54, −12.43] and −16.95 [−17.62, −16.24]. In the pooled strategy, therefore, the spread was sufficient to keep expectancy negative without fees. Bucket-level zero-fee intervals were not estimated and are not implied by this result.

Table 4
Fee sensitivity of pooled taker expectancy

Fee coefficient	Buy NO, ¢/contract	Buy YES, ¢/contract
0.000 (frictionless)	−12.99 [−13.54, −12.43]	−16.95 [−17.62, −16.24]
0.035	−13.99 [−14.54, −13.43]	−17.95 [−18.62, −17.24]
0.070 (standard scenario)	−14.52 [−15.08, −13.96]	−18.57 [−19.24, −17.87]
0.140	−15.54 [−16.09, −14.98]	−19.72 [−20.40, −19.03]

Note. Bracketed values are event-clustered bootstrap 95% CIs based on 2,000 draws.

3.5. Calibration gaps increased at longer horizons

Stratifying the retrospective midpoint gap by time to close at entry reproduced, at the book level, the horizon pattern documented for transaction prices by Page and Clemen (2013). Entries within 24 hours of close showed a gap of −1.89 points (n = 43,324; CI [−2.43, −1.33]), entries between one and seven days −1.73 [−3.22, −0.27], and entries at seven days or more −7.15 [−12.21, −2.23] (n = 1,381; 131 clusters). Calibration quality is not a single number for a venue; it varies with horizon, as capital-cost accounts predict (Grant et al., 2019). The long-horizon stratum is small (131 clusters) and dominated by macroeconomic series, so horizon is entangled with market composition (domain-conditional calibration is documented independently; Le, 2026); Section 3.7 shows the pattern is not reducible to book width.

Table 5
Retrospective midpoint calibration gap by entry horizon

Entry horizon	n	Clusters	Gap, pp	95% CI
< 24 hours	43,324	15,467	−1.89	[−2.43, −1.33]
24 hours–7 days	13,103	4,139	−1.73	[−3.22, −0.27]
≥ 7 days	1,381	131	−7.15	[−12.21, −2.23]

Note. Gap = mean(Y × 100 − p), in percentage points. Intervals use event-clustered bootstrap resampling. Cluster counts do not sum because an event can contribute markets to more than one horizon stratum.

3.6. Displayed coherence violations were not executable in observed scans

Across 130 scans from July 12 through August 24, the display produced a mean of 190.4 ladder-monotonicity violations per scan. None was executable at contemporaneous top-of-book quotes with positive locked profit after the coded fee scenario. Across 65 partition scans, no actionable short-side violation was observed. These are repeated scan-level counts, not unique configurations; the same display relationship may appear in more than one scan. The result is therefore narrow: under the evaluated two-leg rules, no observed display violation survived into an actionable top-of-book arbitrage.

3.7. Robustness: book width and the location of residual miscalibration

Sections 3.1–3.6 report the frozen exhibits. A post-freeze audit prompted one further decomposition, computed from the same frozen data with the same event-clustered procedure (script and outputs accompany the archival materials): midpoint calibration stratified by the entry spread. The motivation is mechanical. A wide, roughly symmetric early book produces a mid-curve midpoint regardless of what the market believes; in this sample, 69.9% of the 40–49-cent midpoint bucket and 55.9% of the 50–59-cent bucket carried entry spreads of 21 cents or more, and the same mid-curve buckets' average entry spreads implied by Table 3 are near 50 cents.

Table 6 reports the stratification. In books with entry spreads of 20 cents or less — 60.6% of the sample — the global midpoint gap was statistically indistinguishable from zero in every stratum, mid-curve buckets included. The favorite–longshot-shaped residual was concentrated in wide books (spreads of 21 cents or more: global gap −4.68 percentage points, 95% CI [−5.57, −3.81]; mid-curve gap −5.66 [−6.53, −4.79]). The horizon pattern of Section 3.5 is not reducible to book width: entries at seven days or more remained miscalibrated within narrow books (−6.37 [−11.97, −1.01]; 90 clusters), though that stratum is small and compositionally distinct. On the paper's own terms, the midpoint of a wide book is a weak price object; much of the apparent mid-curve miscalibration is the signature of uninformative early books rather than a belief anomaly.

The same procedure sharpens two paired contrasts reported above. On identical retrospective markets, the display-minus-midpoint gap difference reduces to the mean of (midpoint − display): −0.46 percentage points (95% CI [−0.55, −0.36]) overall, and −1.18 [−1.42, −0.93] on the 22,136 markets with an explicit historical display value (the reconstruction otherwise substitutes the midpoint, attenuating the contrast toward zero). Prospectively, restricting the display-price analysis to the 583 two-sided markets leaves the tail artifact intact: the 90–99-cent display bucket resolved 81.4% against a 95.0% mean display (n = 86), against 90.4% and 94.8% for the corresponding midpoint bucket — Section 3.1's attenuation is a property of the price object, not of the one-sided-book exclusion.

Table 6

Retrospective midpoint calibration by entry spread (post-freeze robustness)

Spread stratum	n	Clusters	Global gap, pp [95% CI]	Mid-curve gap, pp [95% CI]
1–2¢	13,610	5,994	−0.64 [−1.62, +0.33]	−0.69 [−2.26, +0.86]
3–5¢	12,976	6,663	−0.17 [−1.05, +0.70]	+0.54 [−0.84, +1.94]
6–10¢	5,329	3,072	+0.04 [−1.35, +1.41]	+0.25 [−1.90, +2.39]

Spread stratum	n	Clusters	Global gap, pp [95% CI]	Mid-curve gap, pp [95% CI]
11–20¢	3,135	1,591	+0.93 [−1.39, +3.21]	+1.01 [−2.84, +4.85]
≥21¢	22,758	8,615	−4.68 [−5.57, −3.81]	−5.66 [−6.53, −4.79]

Note. Post-freeze robustness analysis computed from the frozen data (data-freeze-v1) with the event-clustered bootstrap (10,000 draws). Mid-curve = midpoints in [30, 60). Gap = mean(Y × 100 − p), in percentage points.

4. Discussion

The central result survives a skeptical restatement—and, at scale, a reversal of emphasis. The exchange's order-book midpoints were numerically close to calibrated in aggregate, although the sample was large enough to distinguish the global gap and slope from their perfect-calibration values. What miscalibration remained was localized in the middle of the price curve, shaped like the canonical favorite–longshot pattern, larger at longer horizons, amplified at the displayed price, and — in a post-freeze robustness stratification (Section 3.7) — concentrated almost entirely in wide-spread books. Under the standard-fee scenario, all bucket-direction intervals were below zero; pooled estimates remained negative under zero fees; and no displayed coherence violation was executable. Miscalibration and tradable alpha were not the same empirical object.

Three implications follow. First, any forecast evaluation built from market prices should name the price object. A displayed number, last trade, midpoint, and executable quote answer related but nonidentical questions; in these data the choice moved tail calibration by roughly five percentage points. Second, calibration alone does not establish an implementable strategy, and the converse inference—that persistent miscalibration implies someone is leaving money on the table—failed here for a measurable reason: the bid–ask spread exceeded the correction before fees entered. Third, the attenuation from display to midpoint is consistent with an imperfect reference-price layer, but the design does not identify why the displayed number diverged from the book.

The evidence is consistent with, but does not prove, a limits-to-arbitrage account (Shleifer & Vishny, 1997). Spreads, fees, and settlement delay can protect a statistical discrepancy from taker correction. The design measures those wedges; it does not identify the behavioral or institutional mechanism that generated the calibration curve. It also does not contradict platform-wide evidence of strong near-resolution calibration (Kagan & Baiocchi, 2026): the horizon stratification shows how both results can coexist.

4.1. Scope and limitations

The conclusion is strategy-specific. We evaluate top-of-book, hold-to-settlement taker entries under one early-entry policy. Maker-side returns, depth-aware portfolio execution, higher-frequency entry, inventory netting, alternative exit rules, and cross-venue strategies are outside scope. Observing only top-of-book does not weaken a one-contract estimate when the quoted quantity is available; it does mean that larger trades could execute at worse prices. Depth therefore limits scalability rather than strengthening the one-contract result.

The prospective universe was an economics-focused feed before the August 24 expansion and should not be described as the full exchange. The retrospective stratum conditions on an earliest two-sided daily candle, whereas prospective displayed-price calibration can include one-sided books. The windows overlap (770 shared markets), so agreement between strata is a construction check rather than an independent replication. The prospective period is short and regime-specific. The retrospective stratum is dominated by sports and e-sports series and its long-horizon entries by macroeconomic series; category-stratified estimates (cf. Le, 2026) would further separate composition from calibration, and Brier levels should not be compared across strata because they track outcome base rates and market composition. Fee-adjusted bucket conclusions are tied to the standard-fee scenario, while the strongest fee-independent result is the pooled zero-fee sensitivity in Table 4.

5. Conclusion

The bottom line is narrower than “prediction markets are wrong” and more useful. A displayed probability is a forecast; an actionable quote is an offer. On the venue, windows, and entry rules studied, the order-book midpoint was numerically close to calibrated in aggregate; the residual miscalibration was localized, favorite–longshot-shaped, and horizon-dependent; and displayed prices added a tail artifact of their own. Under the standard-fee scenario, every bucket-direction interval was negative, while pooled expectancy remained negative even with fees set to zero. Researchers, journalists, platforms, and traders should say which price object they mean. In this sample, that distinction was not a technical footnote. It was the result.

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Declaration of competing interest

Before this study, the author operated and funded, at small personal scale (less than US$200 total), an automated trading system on the exchange studied. Live trading was retired on June 24, 2026, before construction of the calibration datasets analyzed here. The author has no financial relationship with the exchange.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Claude (Anthropic) and ChatGPT (OpenAI) to assist with research-code development, literature search and verification, content organization, and language editing. The author reviewed and verified the outputs, made all substantive decisions, and accepts full responsibility for the content of the work.

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